Topic 10: Gmail Integration – Test Cases and Results

1. Testing Overview

Workflow: Topic 10 Gmail Integration Workflow

Repository: n8n-gmail-integration

Testing Objective: Verify that n8n can connect to Gmail and send personalized plain-text emails using dynamic customer data.

2. Test Environment

- Platform: n8n Cloud
- Email Service: Gmail
- Authentication: Gmail OAuth2
- Trigger: Manual Trigger
- Data Node: Mock Customer Data
- Email Node: Send Personalized Email

3. Test Data

Field	Test Value
customerName	Charlie Brown
customerEmail	Recipient email configured privately in n8n
Email Type	Text
Subject	Welcome to N8N Course, Charlie Brown!

4. Test Cases and Results

Test ID	Test Scenario	Expected Result	Actual Result	Status

TC-001	Connect Gmail OAuth2 credentials	Gmail account connects successfully	Google sign-in completed and Gmail account connected	PASS
TC-002	Configure Gmail Send operation	Resource is Message and operation is Send	Gmail Send a message action configured	PASS
TC-003	Configure dynamic recipient	Recipient uses customerEmail expression	Recipient expression configured using `{{ $json.customerEmail }}`	PASS
TC-004	Configure personalized subject	Subject includes customerName	Subject preview displayed the personalized name	PASS
TC-005	Configure personalized email body	Body includes customerName and welcome message	Plain-text message preview displayed correctly	PASS
TC-006	Execute Gmail send operation	Email sends successfully and returns message details	Gmail returned message ID, thread ID, and SENT label	PASS
TC-007	Verify email delivery	Email appears in recipient inbox	Recipient inbox checked and delivery verified	PASS
TC-008	Execute complete workflow	All three workflow nodes execute successfully	Complete workflow executed successfully; all three nodes showed green checkmarks	PASS

5. Execution Evidence

Gmail node output:

```
None
[
  {
    "id": "1a1cf8410c2bd6d6",
    "threadId": "1a1cf8410c2bd6d6",
    "labelIds": [
      "SENT"
```

```
      ]
    }
]
```

6. Final Test Summary

Metric	Result
Total test cases	8
Passed	8
Failed	0
Overall result	PASS

7. Conclusion

All eight documented validation checks passed based on the observed workflow configuration, execution output, and email delivery verification.

The workflow successfully demonstrated Gmail OAuth2 authentication and automated personalized plain-text email sending through n8n.